

Spencer Reed Arrasmith

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Driven and capable embedded systems engineer, eager to apply multi-disciplinary problem solving in cutting-edge fields and to empower and educate others through technology and art.

Core Competencies

- C/C++
- Python
- GNU/Linux
- Yocto
- CAD
- Software Architecture
- Circuit Design
- Technical Writing
- Graphic Design

Principal Embedded Software Engineer - BAE Systems SMS

Apr 2020 - Apr 2026

- Software lead on Agile teams for novel phased antenna array systems
- Architected and developed C++ and Rust microservice application suites
- Built and verified Embedded Linux images for custom hardware using Yocto
- Created flexible Python GUI library for system test and control
- Modernized test framework for FAA-certified antenna system
- Wrote 3D render engines abstraction layers for use in capability simulation frameworks
- Led monthly working group for software development ecosystem discussions

Staff Physicist I - Stable Laser Systems

Nov 2014 - Jan 2020

- Developed C++ bare-metal firmware for mixed-signal laser system control
- Designed microcontroller boards implementing UART, SPI, and I2C interfaces
- Implemented embedded touchscreen GUI with custom joystick electronics
- Designed, assembled, and verified ultrastable laser systems
- Managed BOMs, kitting, machine assembly, and electronics production pipelines
- Streamlined and automated workflows for data collection and processing

B.S. Electrical and Computer Engineering - Rice University

Aug 2010 - Jun 2014

- Developed a bluetooth-connected, 3D-printed, panoramic (OpenCV) microscope for Android smartphones
- Completed graduate-level courses in signal processing and filter design
- Gained lab experience with analog circuits, embedded systems, and FPGAs

Personal Projects

Video Game Modding

- Custodian of community development for 2000+ Rocket League modders
- Scripted and recorded 10 hours of video tutorial content for Skillquest.io
- Wrote and hosted a Vuepress-based website with 50,000 words of tutorial content

Hidden Layer Design

- Started an art-meets-technology project with engineer friends
- Brought to life an LED-illuminated, laser cut PCB art piece
- Created an interactive music art installation for the Museum of Boulder

Blender 3D

- Taught an accredited, semester-long course at Rice University
- Gave an internal training course to 20 engineers at Ball Aerospace
- Wrote a publicly-available extension to import Kerbal Space Program vehicles into Blender